

Review

- So far we have discussed linear algebra and matrix theory from a data science perspective
- We will provide some geometric interpretations now
- This section covers the following
 - Vectors
 - Notion of distance
 - Projections
 - Hyperplanes
 - Halfspaces
 - Eigenvalues and eigenvectors



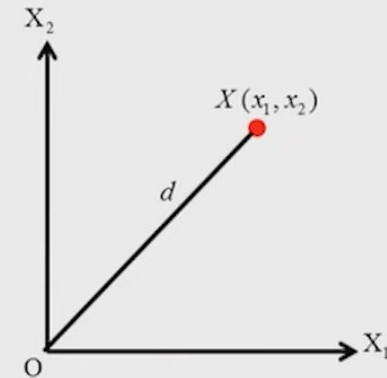
Vectors and lengths

- Consider

$$\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

- $\mathbf{X}$ is a data point in a 2 dimensional plane with x_1 and x_2 as the distances along the $\mathbf{X}_1$ and $\mathbf{X}_2$ axes respectively.
- $\mathbf{X}$ can also be considered as a vector between the origin and the data point
- The length (magnitude) of this vector is

$$d = \sqrt{x_1^2 + x_2^2}$$



Vectors and lengths: Example

- Consider the point $A = (3,4)$ in a two dimensional plane

$$A = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$d = \sqrt{3^2 + 4^2} = 5 \text{ units}$$

- Important: Geometric concepts are easier to visualize in $2D$ or $3D$
- Difficult to do so in the higher dimensions
- However, the fundamental mathematics remain the same irrespective of the dimension of the vector



Vectors and distances

- Consider another example with two points X^1 and X^2

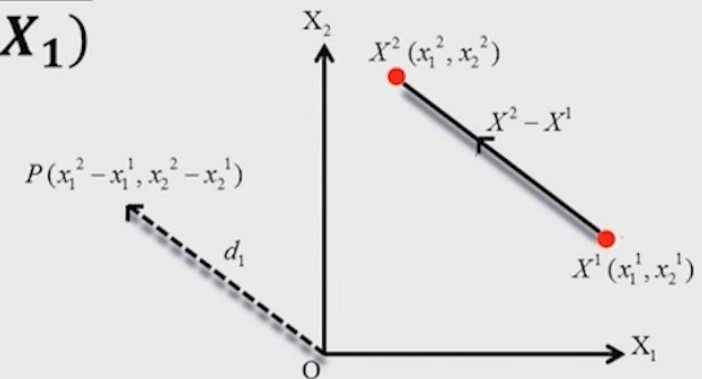
$$\mathbf{X}^1 = \begin{bmatrix} x_1^1 \\ x_2^1 \end{bmatrix} \quad \mathbf{X}^2 = \begin{bmatrix} x_1^2 \\ x_2^2 \end{bmatrix}$$

- The distance between these two can be calculated

$$l = \|\mathbf{X}^2 - \mathbf{X}^1\|_2$$

$$l = \sqrt{(x_1^2 - x_1^1)^2 + (x_2^2 - x_2^1)^2}$$

$$l = \sqrt{(\mathbf{X}^2 - \mathbf{X}^1)^T (\mathbf{X}^2 - \mathbf{X}^1)}$$



Vectors and distances: Example

- What is the distance between points A and B , where A is $(2,7)$ and B is $(5,3)$
- Using the concept of distance introduced before

$$A = \begin{bmatrix} 2 \\ 7 \end{bmatrix}, B = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$
$$l = \sqrt{(5 - 2)^2 + (3 - 7)^2}$$
$$l = 5 \text{ units}$$

Unit vector

- A unit vector is a vector with magnitude 1 (distance from origin)
- Unit vectors are used to define directions in a coordinate system
- Any vector can be written as a product of a unit vector and a scalar magnitude

$$A = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$\text{Magnitude of } A: |A| = \sqrt{3^2 + 4^2} = 5$$

$$\hat{a} = \frac{A}{|A|} = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$$



Orthogonal vectors

- Two vectors are orthogonal to each other when their dot product is 0
- Dot product (scalar product) of two n dimensional vectors A and B

$$A \cdot B = \sum_{i=1}^n a_i b_i$$

- Thus the vectors A and B are orthogonal to each other if and only if

$$A \cdot B = \sum_{i=1}^n a_i b_i = A^T B = 0$$



Orthogonal vectors: Example

- Consider the vectors v_1 and v_2 in 3D space. Identify if they are orthogonal to each other

$$v_1 = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 2 \\ 5 \\ 2 \end{bmatrix}$$

R Code

```
v1=c(1,-2,4)
v2=c(2,5,2)
N=t(v1)%*%v2
```

Console Output

```
> N
      [,1]
[1,]    0
```

- Taking the dot product of the vectors

$$v_1 \cdot v_2 = V_1^T V_2 = [1 \ -2 \ 4] \begin{bmatrix} 2 \\ 5 \\ 2 \end{bmatrix} = 0$$

- Hence, the vectors are orthogonal

Orthonormal vectors

- Orthonormal vectors are orthogonal vectors with unit magnitude
- Example

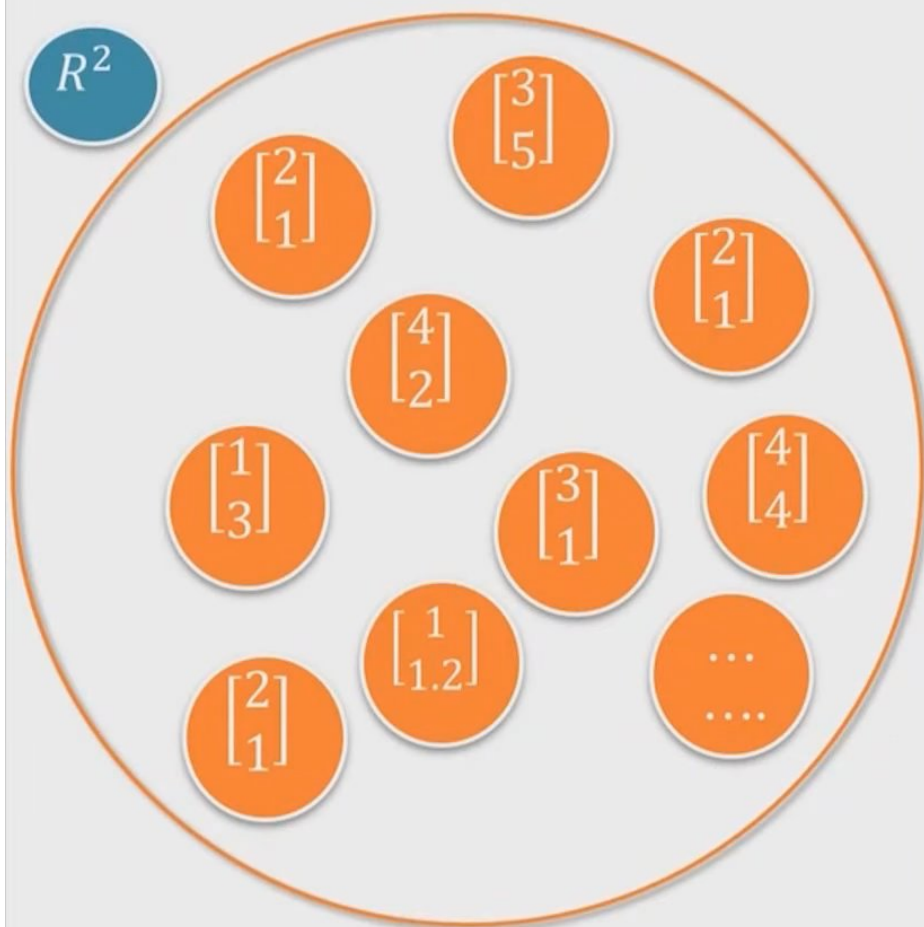
$$v_1 = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} / \sqrt{1^2 + (-2)^2 + 4^2}$$

$$v_2 = \begin{bmatrix} 2 \\ 5 \\ 2 \end{bmatrix} / \sqrt{2^2 + 5^2 + 2^2}$$

- Note that we have taken the vectors from the previous example and converted them into unit vectors by dividing them with their magnitudes.
- All orthonormal vectors are orthogonal



Basis vectors



Let us consider two vectors $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 2v_1 + 1v_2$$

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 4 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 4v_1 + 4v_2$$

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1v_1 + 3v_2$$

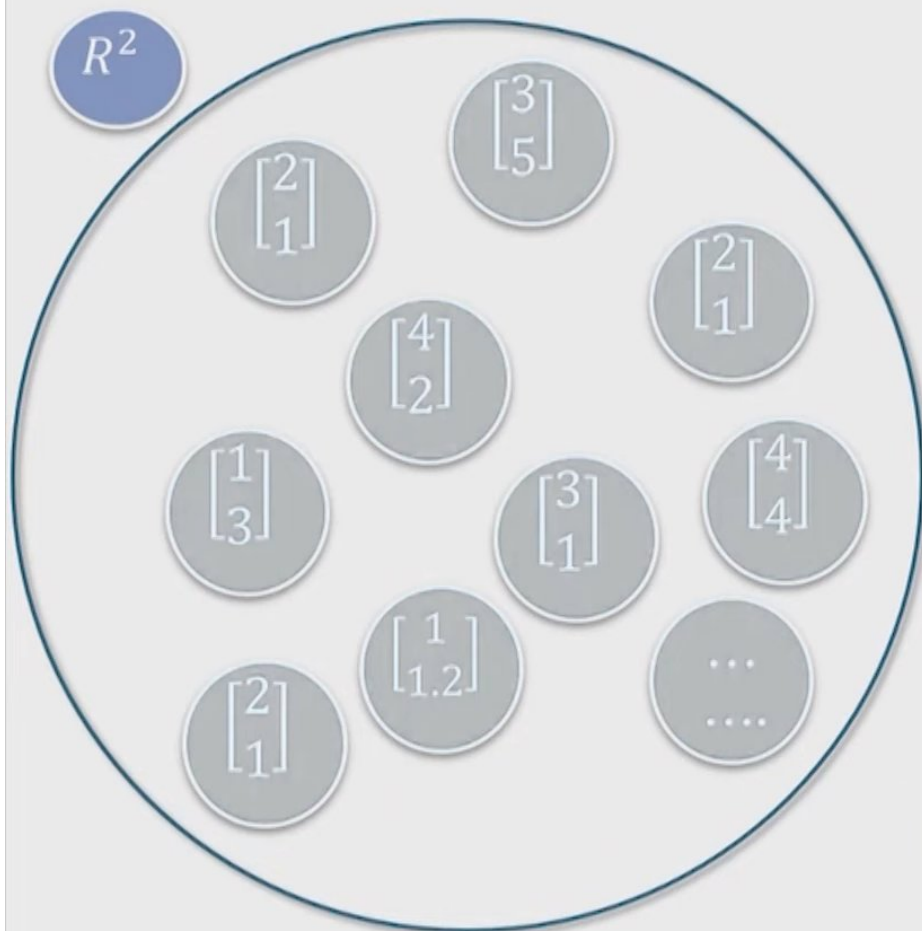
v_1 and v_2 are the basis vectors for R^2

Basis vectors

- Basis vectors are set of vectors that are independent and span the space
- Example:
 - Two vectors $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
 - Can span R^2 and are independent and hence form the basis for the R^2 space.



Basis vectors are not unique



Consider two vectors $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

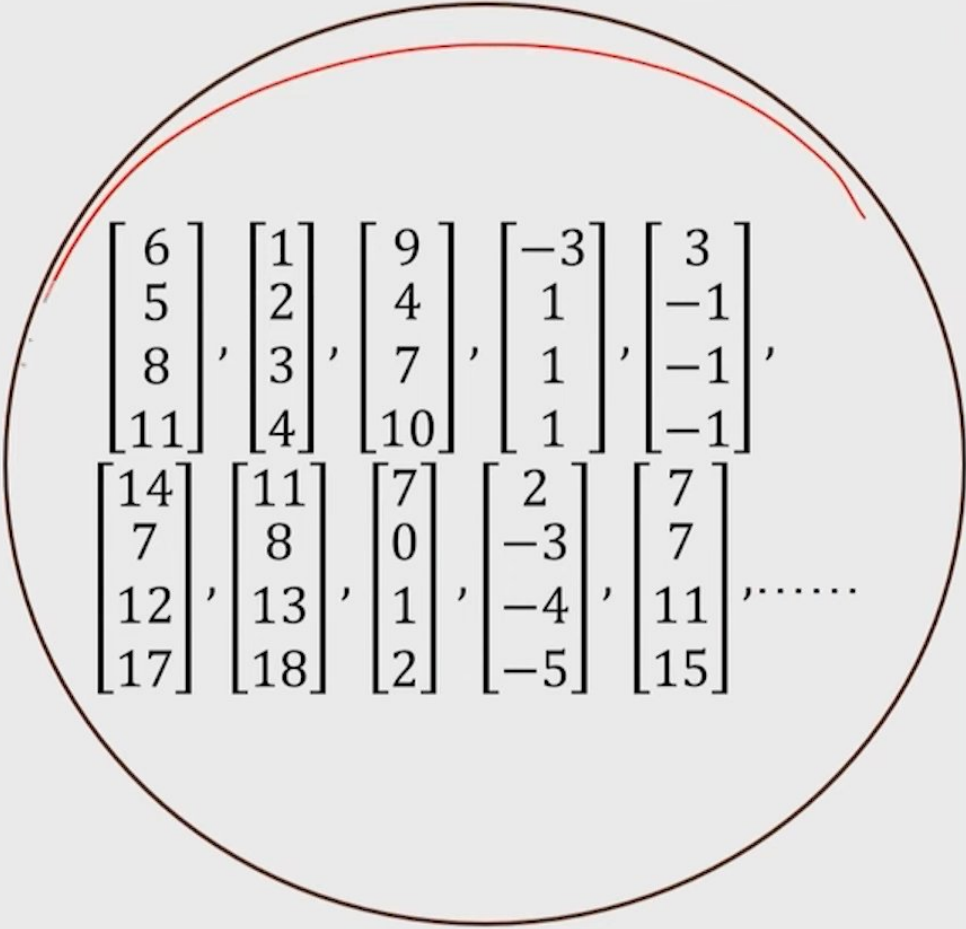
$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = 1.5 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + (+0.5) \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1.5v_1 + (+0.5)v_2$$

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 4v_1 + 0v_2$$

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + (-1) \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 2v_1 + (-1)v_2$$

Hence, this v_1 and v_2 are also basis vectors for R^2

Basis vectors



$$\begin{bmatrix} 6 \\ 5 \\ 8 \\ 11 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 9 \\ 4 \\ 7 \\ 10 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ -1 \\ -1 \end{bmatrix},$$

$$\begin{bmatrix} 14 \\ 7 \\ 12 \\ 17 \end{bmatrix}, \begin{bmatrix} 11 \\ 8 \\ 13 \\ 18 \end{bmatrix}, \begin{bmatrix} 7 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ -4 \\ -5 \end{bmatrix}, \begin{bmatrix} 7 \\ 7 \\ 11 \\ 15 \end{bmatrix}, \dots$$

Consider two vectors $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 4 \\ 1 \\ 2 \\ 3 \end{bmatrix}$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} + 0 \begin{bmatrix} 4 \\ 1 \\ 2 \\ 3 \end{bmatrix} = 1v_1 + 0v_2$$

$$\begin{bmatrix} 7 \\ 7 \\ 11 \\ 15 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} + 1 \begin{bmatrix} 4 \\ 1 \\ 2 \\ 3 \end{bmatrix} = 3v_1 + 1v_2$$

Finding basis vectors

$$\begin{bmatrix} 6 \\ 5 \\ 8 \\ 11 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 9 \\ 4 \\ 7 \\ 10 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ -1 \\ -1 \end{bmatrix},$$

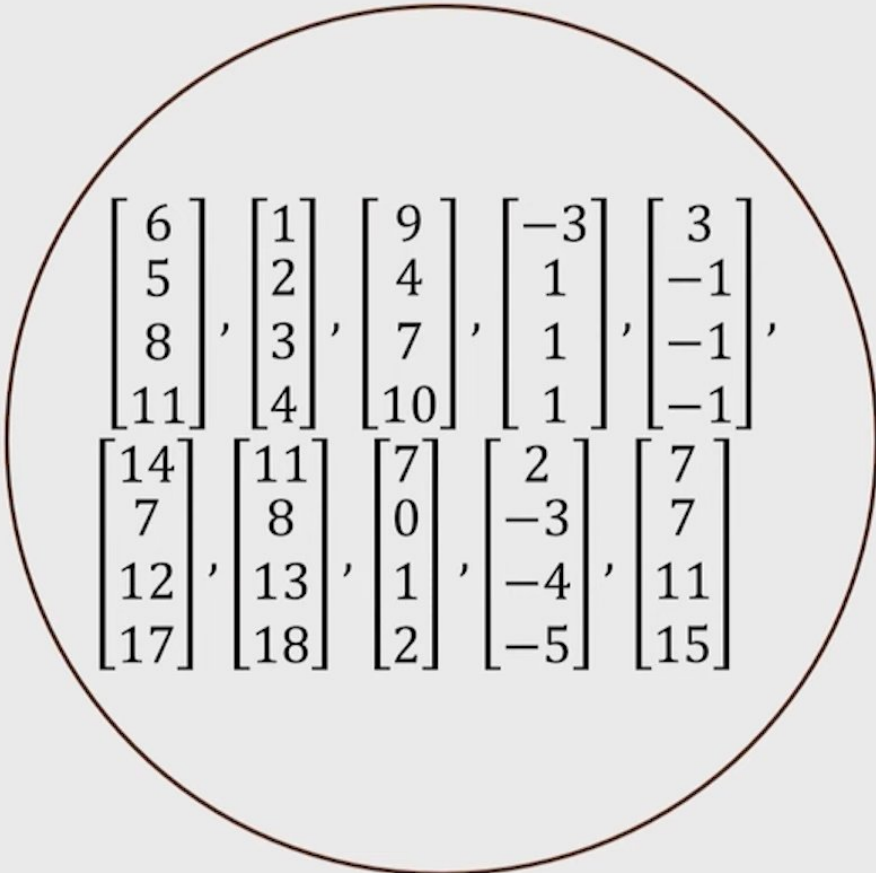
$$\begin{bmatrix} 14 \\ 7 \\ 12 \\ 17 \end{bmatrix}, \begin{bmatrix} 11 \\ 8 \\ 13 \\ 18 \end{bmatrix}, \begin{bmatrix} 7 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ -4 \\ -5 \end{bmatrix}, \begin{bmatrix} 7 \\ 7 \\ 11 \\ 15 \end{bmatrix}$$

To find basis vectors of the given set of vectors, arrange the vectors as shown below

$$\begin{bmatrix} 6 & 1 & 9 & -3 & 3 & 14 & 11 & 7 & 2 & 7 \\ 5 & 2 & 4 & 1 & -1 & 7 & 8 & 0 & -3 & 7 \\ 8 & 3 & 7 & 1 & -1 & 12 & 13 & 1 & -4 & 11 \\ 11 & 4 & 10 & 1 & -1 & 17 & 18 & 2 & -5 & 15 \end{bmatrix}$$



Finding basis vectors



$$\begin{bmatrix} 6 \\ 5 \\ 8 \\ 11 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 9 \\ 4 \\ 7 \\ 10 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ -1 \\ -1 \end{bmatrix},$$

$$\begin{bmatrix} 14 \\ 7 \\ 12 \\ 17 \end{bmatrix}, \begin{bmatrix} 11 \\ 8 \\ 13 \\ 18 \end{bmatrix}, \begin{bmatrix} 7 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ -4 \\ -5 \end{bmatrix}, \begin{bmatrix} 7 \\ 7 \\ 11 \\ 15 \end{bmatrix}$$

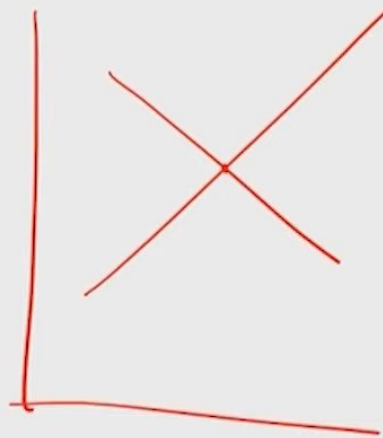
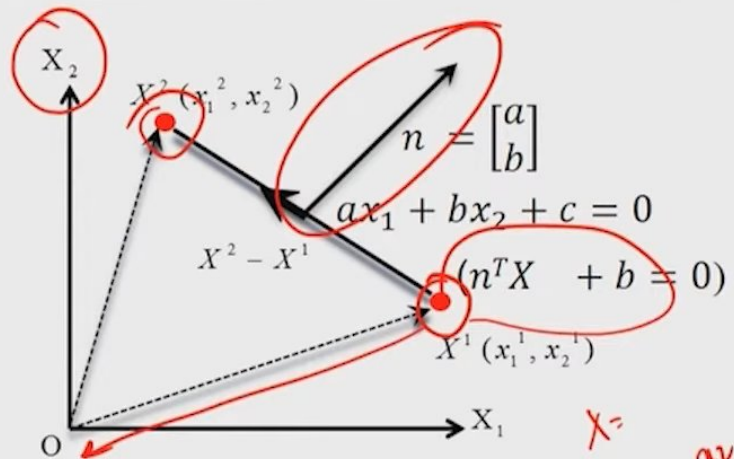
Evaluate the rank of the matrix

$$\begin{bmatrix} 6 & 1 & 9 & -3 & 3 & 14 & 11 & 7 & 2 & 7 \\ 5 & 2 & 4 & 1 & -1 & 7 & 8 & 0 & -3 & 7 \\ 8 & 3 & 7 & 1 & -1 & 12 & 13 & 1 & -4 & 11 \\ 11 & 4 & 10 & 1 & -1 & 17 & 18 & 2 & -5 & 15 \end{bmatrix}$$

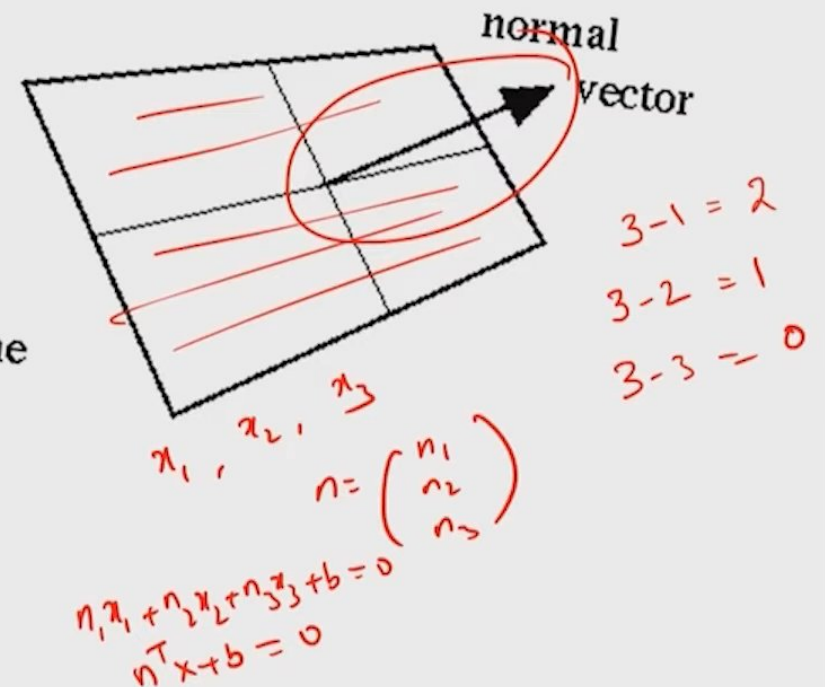
Rank of the matrix is 2

Any two independent columns can be picked from the above matrix as basis vectors

Representation of line and plane



plane



Point X^1 lies on the line $\Rightarrow n^T X^1 + b = 0$ - (1)

Point X^2 lies on the line $\Rightarrow n^T X^2 + b = 0$ - (2)

Subtracting (2) from (1)

$$n^T (X^2 - X^1) = 0.$$

Thus n is perpendicular to $(X^2 - X^1)$

Projections

- We can define the projection ($\hat{X}$) of a vector (X) onto a lower dimension (two dimensions in the picture) mathematically as

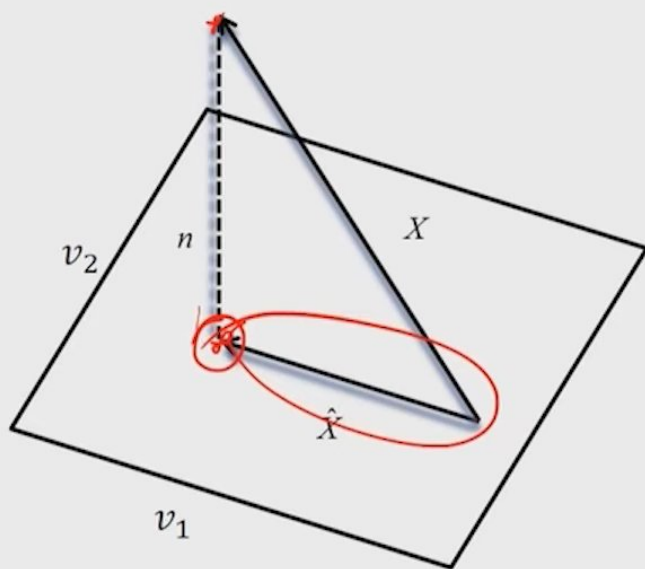
$$\hat{X} = c_1 v_1 + c_2 v_2$$

v_1, v_2

- Using vector addition

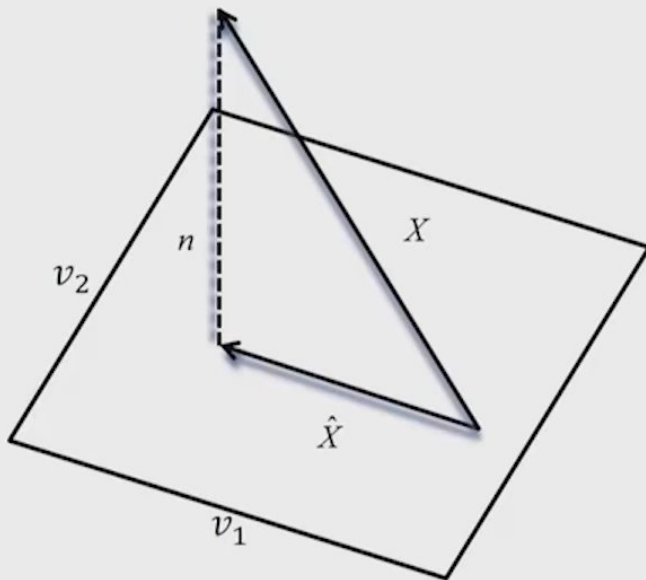
$$X = c_1 v_1 + c_2 v_2 + n$$

$n^T v_1 = 0$
 $n^T v_2 = 0$
 $v_1^T n = 0$
 $v_2^T n = 0$



Projections

- Projections onto general orthogonal directions (two dimensions in this case)



$$\begin{aligned}
 & v_1^T n = 0 \\
 & v_1^T (X - (c_1 v_1 + c_2 v_2)) = 0 \\
 & v_1^T X - c_1 v_1^T v_1 = 0 \\
 & \hat{X} = \frac{v_1^T X}{v_1^T v_1} v_1 + \frac{v_2^T X}{v_2^T v_2} v_2
 \end{aligned}$$

Handwritten notes in red:

- $-c_2 v_1^T v_2$ (circled)
- $v_2^T n = 0$
- $c_1 = \frac{v_1^T X}{v_1^T v_1}$
- $c_2 = \frac{v_2^T X}{v_2^T v_2}$

Projections: Example

$$X = [1 \ 2 \ 3]^T$$

- Projecting this vector onto the space spanned by the vectors $v_1 = [1 \ -1 \ -2]^T$ and $v_2 = [2 \ 0 \ 1]^T$
- Thus, finding the projection onto the plane defined by v_1 and v_2 is

$$\hat{X} = \frac{[1 \ 2 \ 3] \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}}{6} \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix} + \frac{[1 \ 2 \ 3] \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}}{5} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

Projections: Example

$$\hat{X} = \frac{-7}{6} \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

$$\hat{X} = \begin{bmatrix} 5/6 \\ 7/6 \\ 20/6 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$



Projection -Generalization

- Projections onto general directions
- Consider the problem of projection of X onto a space spanned k linearly independent vectors

$$\hat{X} = \sum_{j=1}^k c_j v_j$$

$$\hat{X} = [v_1 \quad \dots \quad v_k] \begin{bmatrix} c_1 \\ \vdots \\ c_k \end{bmatrix}$$

$$\hat{X} = Vc$$



Projection -Generalization

- Using orthogonality idea

$$X = \hat{X} + n$$
$$n = X - \hat{X}$$

$$V^T(X - \hat{X}) = V^T(X - Vc) = 0$$

$$V^T X - V^T V c = 0$$

$$c = (V^T V)^{-1} V^T X$$

$$\hat{X} = V(V^T V)^{-1} V^T X$$

